

Data Analyst Capstone Project

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OUTLINE



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EXECUTIVE SUMMARY



- Washington DC is the strongest job market, generating roughly 5,300 postings — more than any other city and nearly 40% above the next closest market.
- Salary and demand don't move together: Swift pays the highest average salary (~\$130K), but the highest-volume job hubs (DC, Detroit, Seattle, NY) are not primarily Swift/iOS markets, so top-paying skills and top-hiring cities are separate opportunities. Sub
 - Python, C++, and JavaScript form a tight second tier, each averaging \$110K–\$115K.
 - PostgreSQL is now the single most in-demand database, both currently and looking ahead to next year.
 - Younger developers (18–24 and under 18) show disproportionately higher interest in emerging languages compared to established databases.
- The developer population is concentrated and young — 25–34 year-olds alone make up 41.3% of respondents, and 25–44 year-olds together account for nearly 70%.
- Cloud infrastructure is a two-horse race: AWS and Microsoft Azure dominate current and future usage, though Google Cloud and DigitalOcean are gaining ground as multi-cloud adoption grows.
- Spring Boot leads web framework adoption today and is expected to remain the top choice next year, with React and ASP.NET Core as strong runners-up.



INTRODUCTION



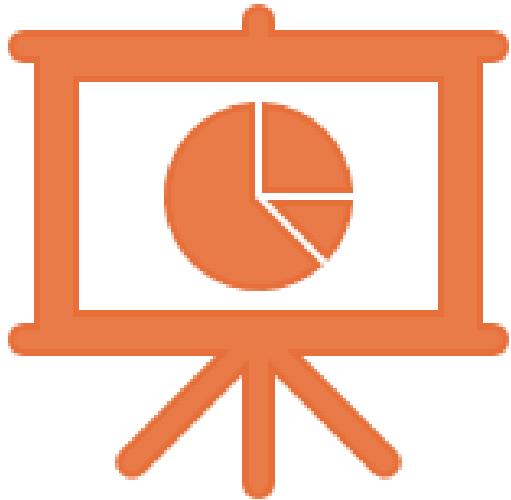
- This report analyzes job posting volume, in-demand programming languages, salary data, and database/technology trends to identify where opportunity is concentrated in the current data/tech job market.
- The intended audience is job seekers, hiring managers, and workforce/education planners who need an evidence-based view of which skills and locations offer the strongest returns.
- The analysis combines job-posting API data, web-scraped salary data, and developer survey data (technology usage and demographics) to triangulate demand, compensation, and future direction.
- The value of this analysis is in turning three separate datasets into one coherent picture of "where the jobs are, what they pay, and what's coming next."
 - For job seekers, it clarifies which languages/skills to prioritize learning.
 - For employers and planners, it flags which platforms and databases are becoming the industry standard versus which are declining.

METHODOLOGY



- Job posting counts by city were collected via a Job Postings API and stored in job-postings.xlsx.
- Programming language salary data was collected via web scraping and stored in popular-languages.csv.
- Programming language and database usage/preference data came from a large-scale developer survey dataset, segmented by age, employment status, and country.
- Data wrangling steps included cleaning inconsistent city/language labels, removing null and duplicate survey responses, and aggregating "Worked With" vs. "Want to Work With" counts for trend comparisons.
 - Categorical fields (age brackets, education level, employment type) were standardized before building demographic charts.
 - Charts were sorted in descending order (by job count or salary) to make rankings immediately readable.

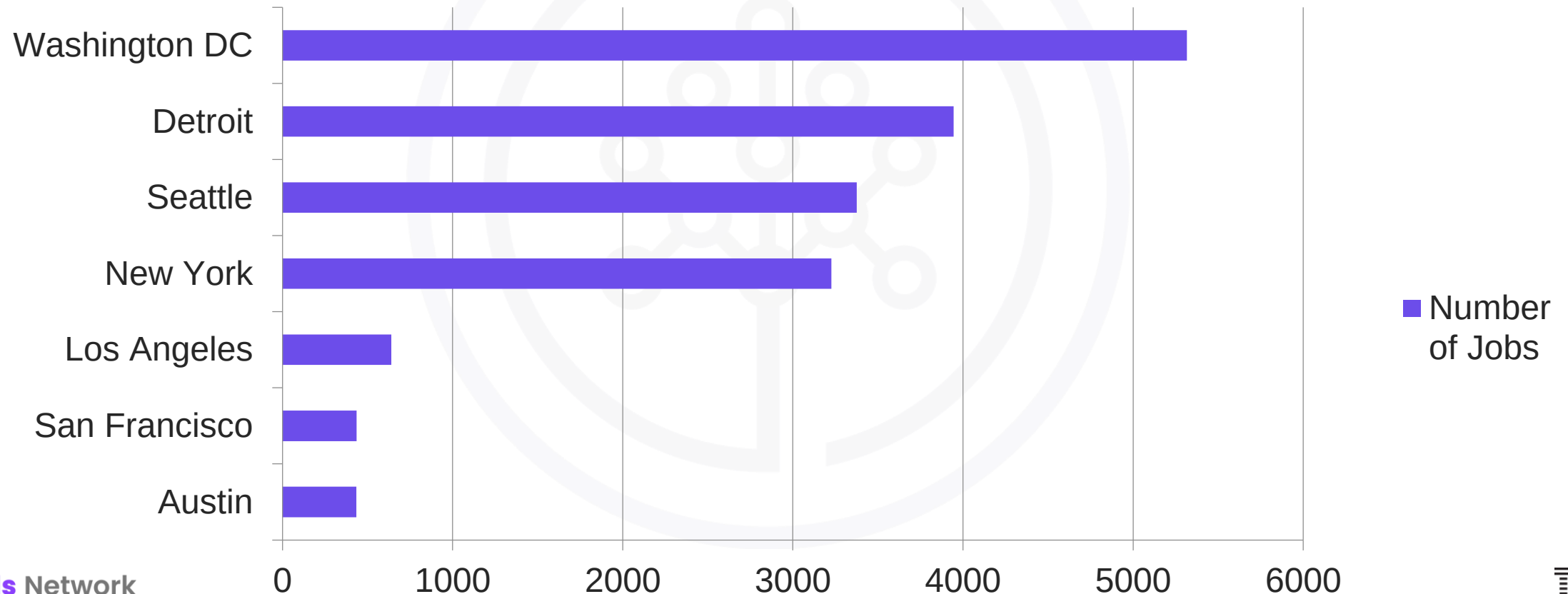
Results



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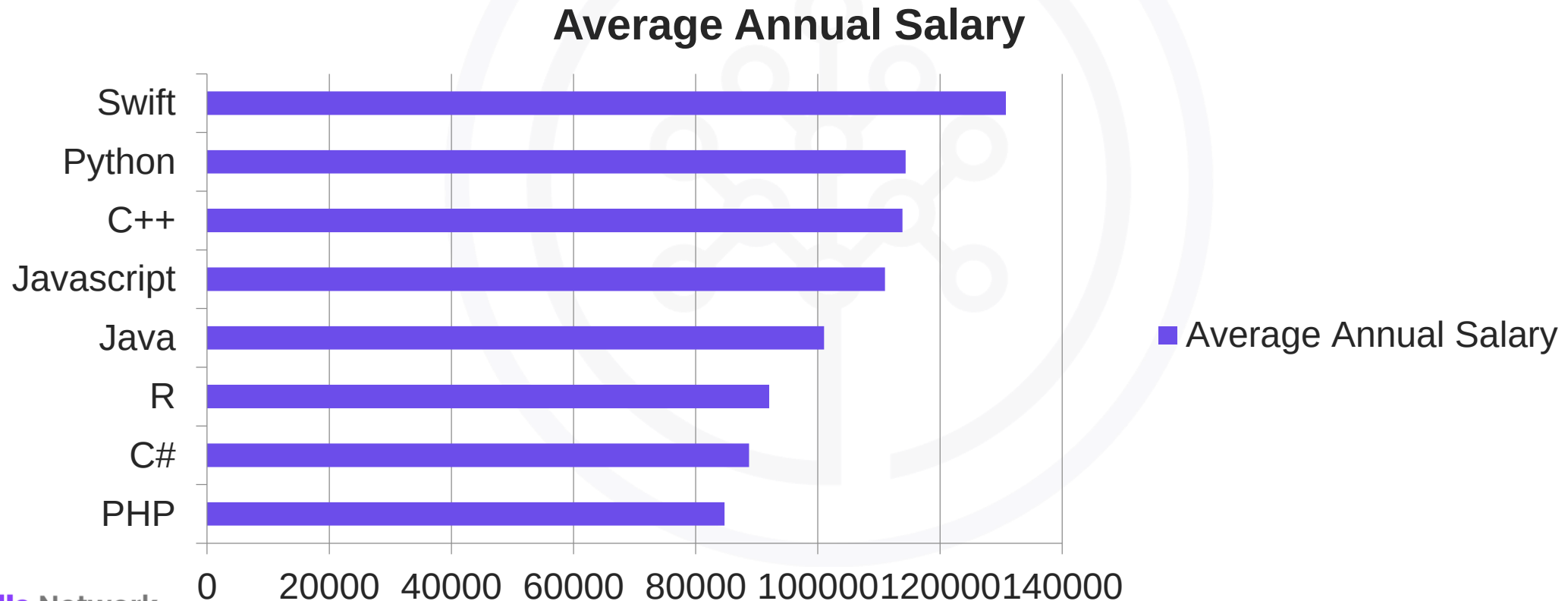
JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.

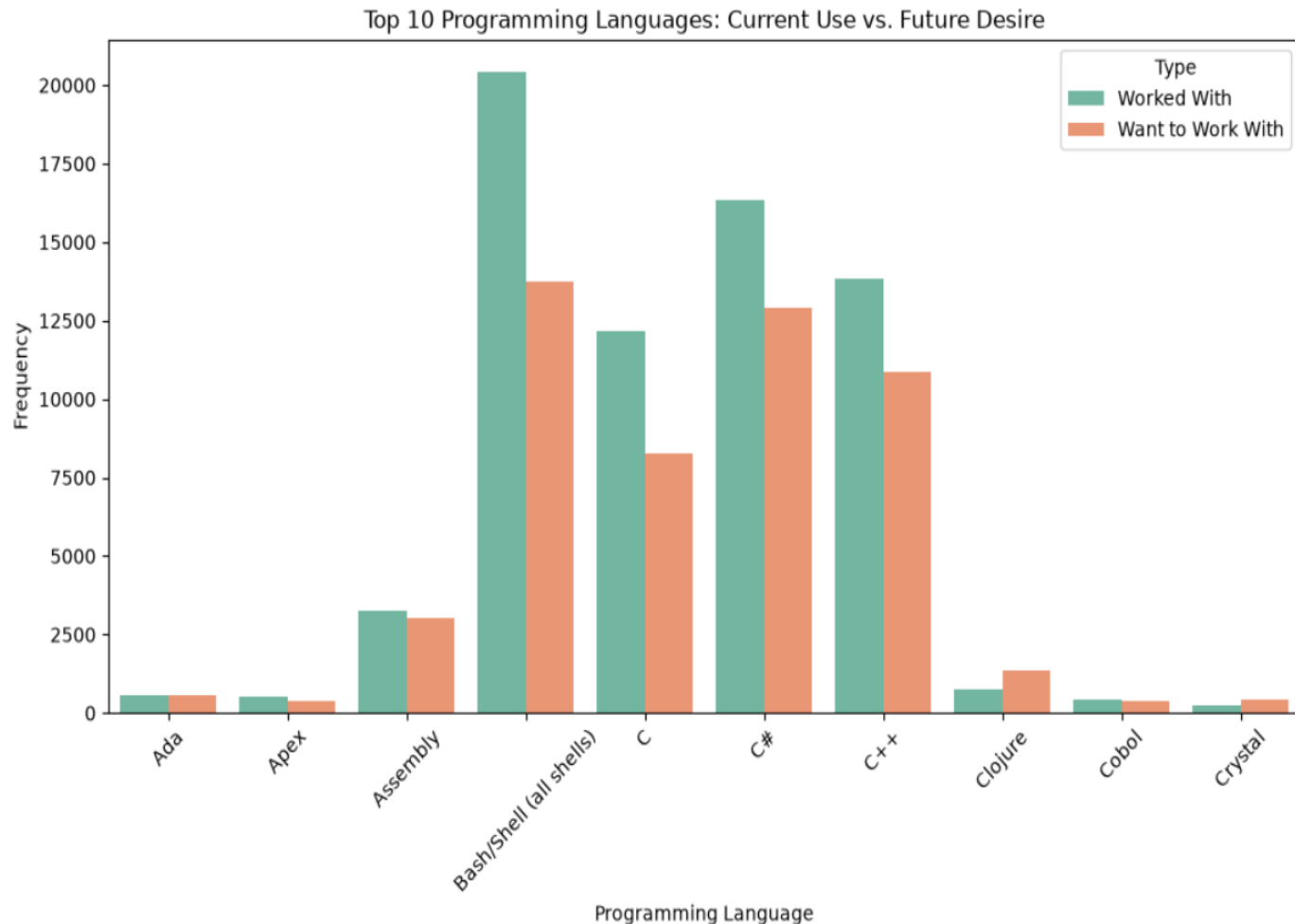


POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.



PROGRAMMING LANGUAGE TRENDS



Bash/Shell, C#, and C++ are the most-used languages today, each with well over 10,000 respondents reporting current use. However, "Want to Work With" counts are lower than "Worked With" counts for nearly every major language (Bash/Shell, C, C#, C++), signaling that developers are using these languages out of necessity more than preference. Niche languages like Ada, Apex, Clojure, Cobol, and Crystal have small but stable footprints, with Clojure and Crystal actually showing slightly higher future interest than current use — a sign of a small but growing niche following.



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Every top-5 current language shows a drop-off between "Worked With" and "Want to Work With," indicating skill fatigue or legacy-code obligation rather than enthusiasm.
- C++ and C# retain the smallest relative decline, suggesting durable long-term demand.
- Smaller/niche languages (Clojure, Crystal) are among the few with rising future interest, hinting at emerging developer preference away from legacy shell-based work.

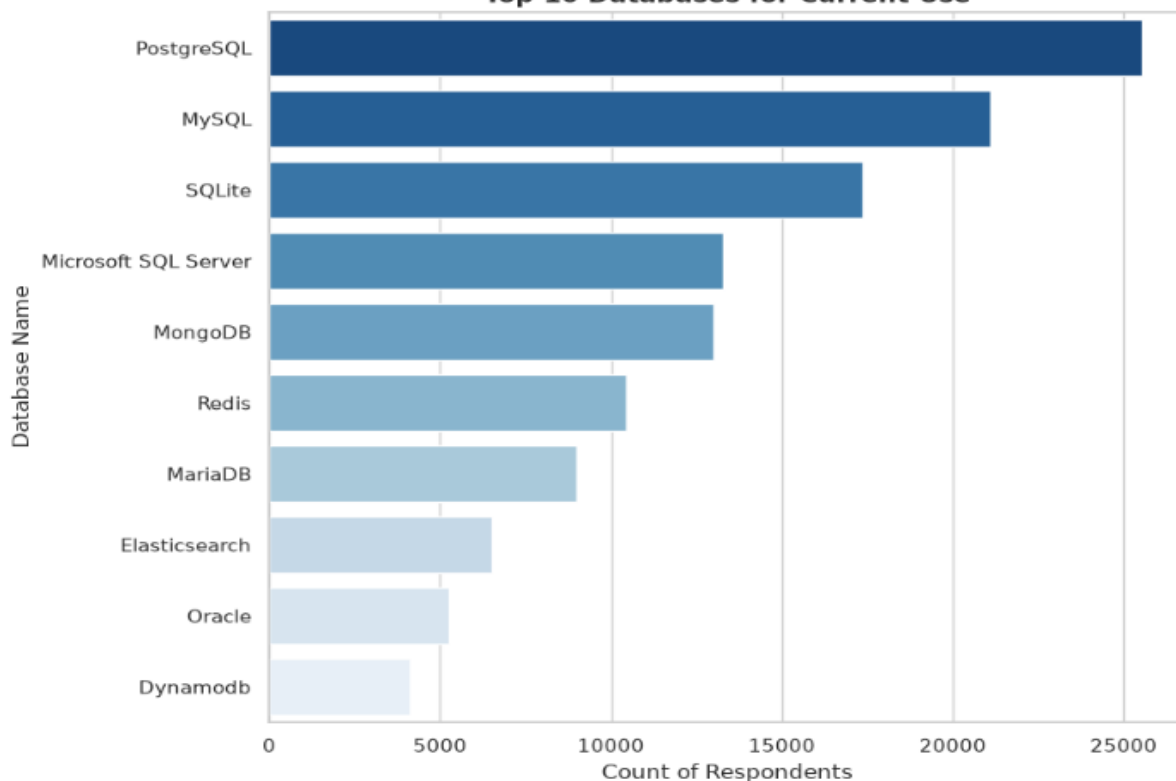
Implications

- Training programs should not assume current top languages will retain the same demand share going forward.
- Employers relying heavily on Bash/Shell or C-family stacks may face growing difficulty attracting new talent unless roles emphasize modernization.
- Watching niche-language growth (even from a small base) can provide early signal on where developer sentiment is heading.

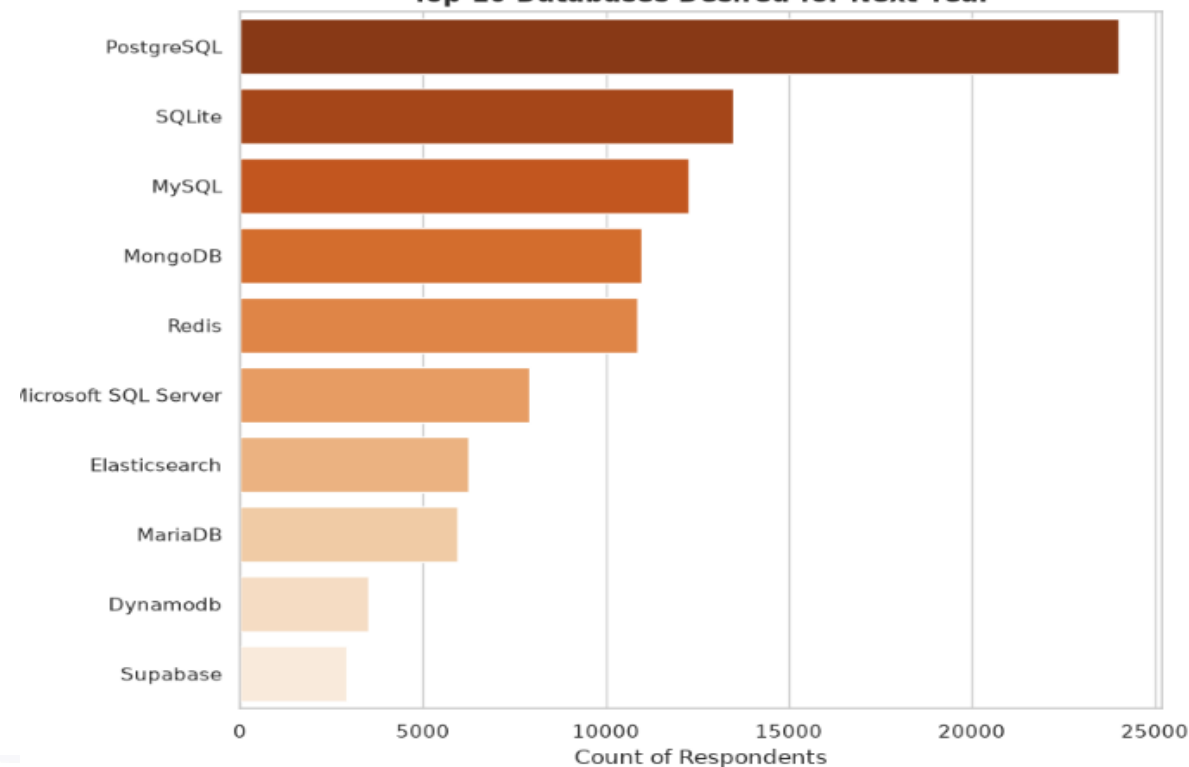
DATABASE TRENDS

PostgreSQL is the clear leader both in current use (~25,000 respondents) and future interest (~24,000), confirming its position as the industry-standard relational database. MySQL and SQLite round out the top 3 today, but their order shifts: SQLite overtakes MySQL in future demand, moving from 3rd to 2nd place. MongoDB and Redis maintain consistent mid-tier positions in both charts. Notably, Supabase appears as a new entrant in the "desired next year" ranking (~top 10) despite not appearing in current use, signaling fast-emerging adoption. Oracle and traditional enterprise databases fall out of the top 10 for future interest entirely.

Top 10 Databases for Current Use



Top 10 Databases Desired for Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL has effectively become the default choice for both current and future database work.
- Lightweight/serverless options (SQLite, Supabase) are gaining share at the expense of heavier enterprise systems (Oracle, MS SQL Server).
- MongoDB and Redis show stable, unchanged demand — neither growing sharply nor declining.

Implications

- Teams standardizing on new infrastructure should default to PostgreSQL unless there's a specific reason not to.
- The rise of Supabase suggests growing developer preference for managed, serverless backend solutions — worth monitoring for hiring and tooling decisions.
- Legacy enterprise database skills (Oracle) may see shrinking demand, which has workforce-retraining implications for teams built around those systems.

DASHBOARD

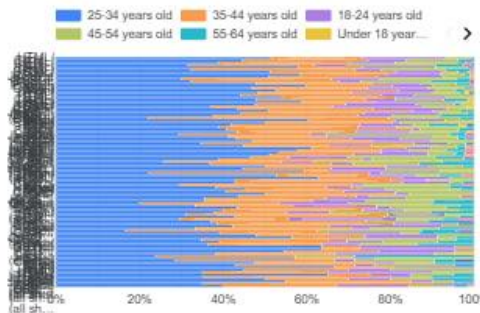


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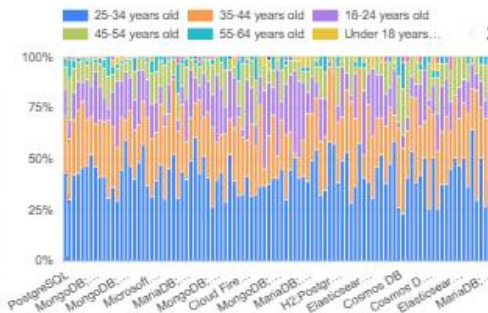
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

Top 10 Languages Used



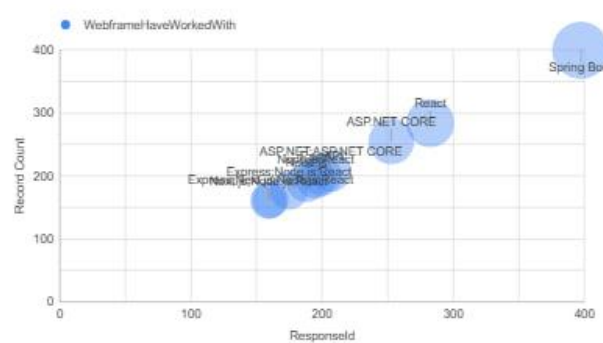
Top 10 Databases Used



Top 10 Platforms Used



Top 10 Web Frameworks Used



1. Languages and Databases

- **Dominant Demographics:** Developers aged 25–34 years old (blue segment) represent the largest user base across almost all primary languages and databases.
- **Stable Core Technologies:** Traditional powerhouse tools (like the top items in both charts) maintain a heavily distributed, massive volume of users compared to niche alternatives.

2. Cloud and Infrastructure (Platforms)

- **Market Leaders:** Amazon Web Services (AWS) and Microsoft Azure clearly dominate cloud infrastructure, appearing as the largest text elements in the Word Cloud.
- **Secondary Ecosystems:** Google Cloud, Cloudflare, and DigitalOcean follow as strong secondary players, while specialized tools (like Netlify and Heroku) occupy smaller, specific market shares.

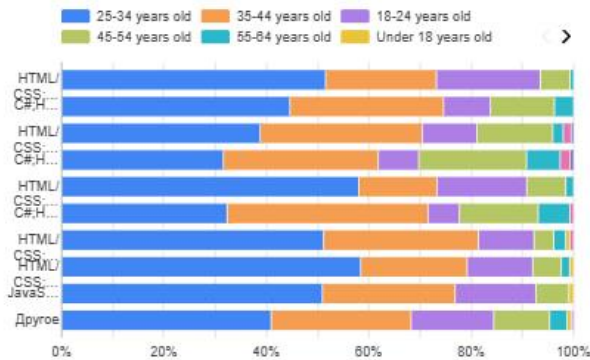
3. Web Frameworks Market Share

- **The Leader:** Spring Boot stands out as the most widely used web framework, positioned high and to the right with a massive volume bubble.
- **The Contenders:** ASP.NET Core and React form a close secondary cluster with substantial user counts, while options like Node.js, Express, and Next.js sit closer together in a highly competitive mid-tier group.

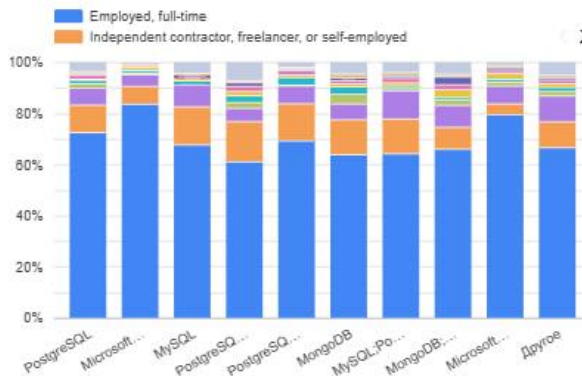
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trends

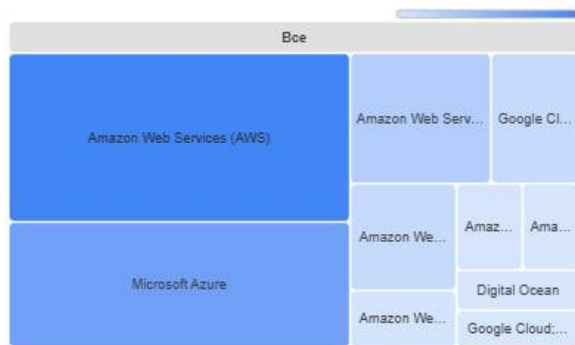
Top 10 Languages Desired Next Year



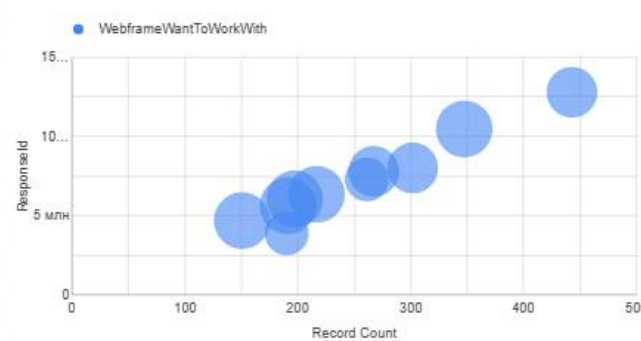
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



Top 10 Desired Web Frameworks



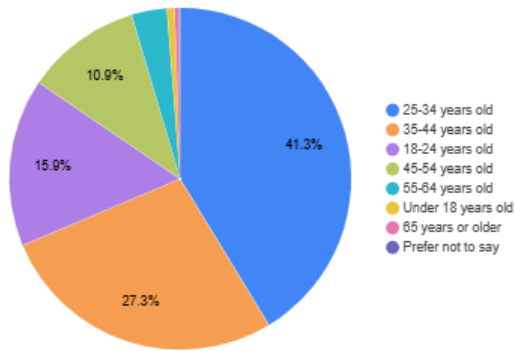
- **Generational Shifts in Language Interest:** Younger developers (groups aged 18–24 and under 18) show a significantly higher enthusiasm for learning next-generation programming languages compared to traditional databases.
- **Database Stability:** The future database market remains highly stable and inert, driven mostly by Employed, full-time specialists. PostgreSQL, MySQL, and MongoDB continue to be the top choices that developers want to work with next year.
- **Multi-Cloud Momentum:** While AWS and Microsoft Azure retain the largest share of future interest on the Tree Map, the footprint for Google Cloud and DigitalOcean has expanded, indicating a growing industry pivot toward multi-cloud architectures.
- **Web Framework Consolidation:** Future developer interest continues to aggregate around Spring Boot, React, and ASP.NET Core. Their high record counts solidify these frameworks as the definite industry standards for the upcoming year.



DASHBOARD TAB 3: Demographics

Demographics

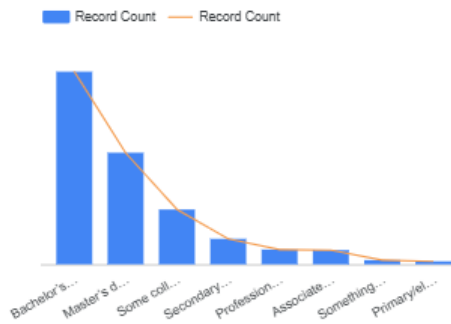
Respondents by Age



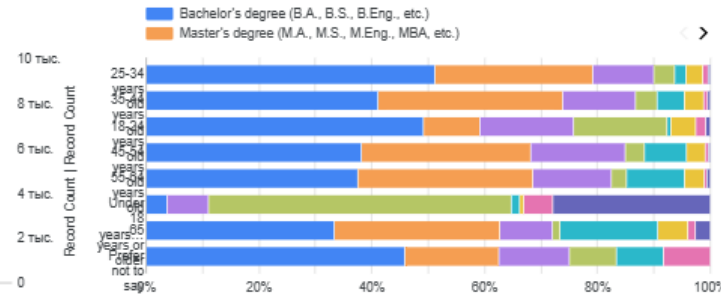
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



- **Age Distribution:** The global developer community is overwhelmingly young and career-active. The 25–34 years old cohort makes up the absolute majority at 41.3%, followed closely by the 35–44 years old demographic at 27.3%. Combined, they account for nearly 70% of all respondents.
- **Geographical Spread:** The map chart shows a highly globalized footprint with dense clusters of respondents across North America, Europe, and India, reflecting the primary hubs of contemporary tech industry talent.
- **Educational Attainment:** Higher education is a prominent feature of the community. A Bachelor's degree is the most common qualification, followed sequentially by a Master's degree. Together, university-level degrees account for the vast majority of the respondent distribution.
- **Education by Generational Cohort:** When breaking down age by educational classification, the data demonstrates that a Bachelor's degree (blue segment) remains highly prominent across younger entry-level developers (e.g., 18–24 years old). Meanwhile, Master's degrees (orange segment) expand noticeably in older brackets (35–44 and 45–54 years old), highlighting academic advancement alongside career progression.

DISCUSSION



- **A young, concentrated developer base is driving the trends**
Nearly 70% of respondents are between 25–44 years old, clustered heavily in North America, Europe, and India. This isn't a niche sample — it's mainstream, career-active developers, which means the technology preferences in Tabs 1 and 2 reflect where the core of the industry is heading, not just early adopters.
- **Current usage and future intent are remarkably aligned — consolidation, not disruption**
Comparing Tab 1 to Tab 2, the same names dominate both: PostgreSQL/MySQL/MongoDB for databases, AWS/Azure for cloud, Spring Boot/React/ASP.NET Core for frameworks. There's no major shake-up coming — the market is consolidating around a stable set of standards rather than shifting to something new.
- **Multi-cloud is the one area actually moving**
The clearest change between "current" and "desired next year" is the growing footprint of Google Cloud and DigitalOcean alongside AWS/Azure. This is the closest thing to a genuine trend shift in the data — teams hedging away from single-cloud dependency.
- **Education level tracks career stage, not just preference**
Bachelor's degrees dominate among younger respondents (18–24), while Master's degrees become more common in the 35–44 and 45–54 brackets — suggesting many developers pursue advanced degrees *after* entering the workforce rather than before, which has implications for how "qualification" should be read in hiring data.
- **Demographic concentration amplifies the technology findings**
Because the respondent base is so concentrated by age and geography, the dashboard's "top technology" rankings likely represent an echo chamber of similar-background developers rather than a fully global signal — a caveat worth flagging when using this data to make broad industry claims.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Job demand is geographically concentrated — Washington DC, Detroit, Seattle, and New York together account for the large majority of postings, while West Coast hubs (LA, SF) and Austin lag well behind despite their tech reputations.
- Salary leadership (Swift, Python, C++) doesn't map directly onto the highest-volume languages in daily use (Bash/Shell, C#, C++), meaning the best-paid skill isn't always the most commonly required one.
- The technology stack is consolidating around a small set of standards — PostgreSQL for databases, AWS/Azure for cloud, and Spring Boot/React/ASP.NET Core for web frameworks — across both current use and future intent.

Implications

- Job seekers optimizing purely for salary (e.g., learning Swift) should verify that target cities/roles actually demand that skill, since high pay and high posting volume don't always overlap.
- Organizations should prioritize consolidating around the "winning" standards (PostgreSQL, AWS/Azure, Spring Boot) to align with where both current usage and future hiring pools are heading.
- The rising engagement from younger developers (18–24, under 18) in emerging tools signals where the future talent pipeline's preferences are shifting, which should inform long-term tooling decisions.



CONCLUSION



- Job demand is concentrated in a handful of metro areas (Washington DC, Detroit, Seattle, New York), making location strategy a major factor in job search outcomes.
- Compensation leaders (Swift, Python, C++) and usage leaders (Bash/Shell, C#, C++) are not identical, so skill strategy should weigh both salary and market demand.
- The technology ecosystem is consolidating around a clear set of standards: PostgreSQL, AWS/Azure, and Spring Boot/React/ASP.NET Core.
- The developer population skews young (70% under 45), with younger cohorts driving interest in next-generation languages and platforms — a leading indicator of where the industry is headed.

APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.